

MAT 426, Class 12: The QR Factorization

QR FACTORIZATION *If A is an $m \times n$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an $m \times n$ matrix whose columns form an orthonormal basis for $\text{Col}(A)$ and R is an $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.*

Example 1. Find a QR factorization of $A = \begin{bmatrix} 1 & 2 & 2 \\ -1 & 1 & 2 \\ -1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix}$.

Definition. Let A and B be $n \times n$ matrices.

- The matrix A is **similar** to B if there exists an _____ $n \times n$ matrix P such that _____.
- The matrix B is **orthogonally similar** to B if there exists an _____ $n \times n$ matrix Q such that _____.

THEOREM Let A and B be $n \times n$ matrices that are similar. Then,

- (a) $\det(A) = \det(B)$
- (b) A is invertible if and only if B is invertible
- (c) A and B have the same rank
- (d) A and B have the same characteristic polynomial.
- (e) A and B have the same eigenvalues
- (f) A^m is similar to B^m for all integers $m \geq 0$
- (g) If A is invertible, then A^m is similar to B^m for all integers m .

Proof of (a), (d), and (e).

QR Algorithm for Computing Eigenvalues

QR ITERATION FOR COMPUTING EIGENVALUES

Input:

Outputs:

Step 1: Set $A_0 = A$

Step 2: For $k = 1, 2, \dots$,

- Find the _____ of A_{k-1}
- Compute $A_k = \text{_____}$.

THEOREM The matrix A_k in the QR Iteration is orthogonally similar to A_{k-1} . Furthermore, the matrices A_k are all orthogonally similar to A .

Question: Why is it important that the matrices A_k are orthogonally similar to A ?

Proof.

Example 2. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

- (a) Determine the eigenvalues of A .

(b) Compute four steps of the QR iteration for the matrix A .

CONVERGENCE OF QR ITERATION Suppose that A is an $n \times n$ matrix that has eigenvalues $\lambda_1, \dots, \lambda_n$ with $|\lambda_1| > |\lambda_2| > \dots > |\lambda_n|$. Let X denote the matrix whose i th column is an eigenvector of A corresponding to λ_i and suppose that X^{-1} has an LU decomposition. Then, the subdiagonal entries of the matrices A_k tend to zero and for $i = 1, 2, \dots, n$, $(A_k)_{ii} \rightarrow \lambda_i$.

Idea of Proof:

- Existence of X^{-1} :

- Orthogonal Similarity to A :

- Diagonalization of A :

- Sufficiency of showing that $\lim_{k \rightarrow \infty} P_k = Q$:

- Why does $\lim_{k \rightarrow \infty} P_k = Q$?

- Why does $\lim_{k \rightarrow \infty} D^k L D^{-k} = I$